Finlay Cooper

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Summary

Software engineer who ships production systems end to end, solo when needed. Sole engineer on a compliance platform running at all 3 of BroadGrain's Canadian plants. Builds with AI agents by default: sets the architecture, schemas, and tests, runs agents in parallel, ships what passes.

Experience

Software Developer Intern, BroadGrain Commodities | TypeScript · React Native · Node.js · MySQL Summer 2025

- Sole engineer on a compliance platform live at all 3 Canadian plants; **cut reporting time 75%**.
- Scoped with plant managers at each site and shipped iterative releases into daily production.
- Built with **parallel AI agents** working from project context files and schemas defined up front.
- Designed a **schema-driven backend factory**, collapsing **100+ hand-written files into 3**.
- Merged legacy .NET web and MAUI mobile apps into **one React Native app** for iOS, Android, and web.
- Migrated all production data to a normalized schema, using **MCP tooling** to derive and map it.
- Replaced paper across 8 compliance categories with **10 PDF reports** auto-delivered to SharePoint.

Technical Lead, GTA Management System | Next.js · TypeScript · Express · Prisma · MySQL 2025–2026

- Led a 4-person team on the TA-hiring platform now used by **100+ people** in Western ECE.
- Designed and built the **matching algorithm**: scores every student-course pair, fills TA hours in waves.
- Set the architecture and **rebuilt the backend in 3 days with AI agents**: 59 files, 100+ endpoints.
- Locked down **14 API modules** with JWT auth, role and ownership checks, and Zod validation.
- Wrote the **three-role test harness** that exercises every endpoint and gates each change.
- Built **dry-run matching**: the committee previews and tunes weights without touching approved picks.
- Built the **typed API client**, reusable CRUD components, and drag-and-drop matching board.
- Deployed to **Railway and Netlify** and wrote the API, auth, and component docs the team built on.

Freelance Developer, Soi Ram | Next.js · TypeScript · Tailwind · Toast POS June 2026

- Designed, built, and launched **soiram.ca**, from brand identity to deployment.
- Shipped design changes **live on client calls** with AI agents, building her vision in real time.
- Integrated **Toast POS direct ordering** alongside UberEats and DoorDash to cut commission fees.
- Built a **retro 80s brand**, data-driven menu, dynamic theming, and mobile-first SEO layout.

Projects

Fracture Protocol, 2-Person Roguelike | Unity · C# April 2026

- Had a **full playable version in 2 days with AI agents**, writing two-thirds of the C#.
- Built all **3 boss fights** and the elite roster with coroutine AI and HP-based phase transitions.
- Built **32 modifiers in 5 rarity tiers** with weighted drops, stack caps, and diminishing returns.
- Traced a **web-build-only combat bug** to Unity stripping a shader, and shipped a build-safe fallback.

Skills

Languages & Frameworks: TypeScript/JavaScript, Java, C#, Python, Bash, SQL; React, React Native/Expo, Next.js, Node.js, Express.js, Tailwind CSS, Unity; Linux

Spoken Languages: English (Native), French (Fluent)

Education

Western University, London ON June 2026

Bachelor of Engineering, Software Engineering | **Dean's Honour List**